

EVIDENCE-BASED PERFORMANCE ASSESSMENT OF THE UNITREE L1 3D LiDAR

A targeted evidence synthesis and secondary analysis for indoor mobile robotics

Author	Pascual Asio Serrano
Context	Research internship, ISR-UC / Human-Centered Mobile Robotics Laboratory, University of Coimbra
Date	6 August 2026

STUDY STATUS

This document does not report new measurements from the Coimbra sensor. It re-analyses published numerical results and separates manufacturer specifications, independent observations, and author-derived inferences.

Abstract

This study addresses the question: under which conditions does the Unitree L1 provide measurements and spatial coverage sufficiently stable for perception and mapping on an indoor mobile robot? A targeted synthesis of primary sources was combined with a new secondary analysis of published Unitree L1 odometry results. The manufacturer specifies a 360 x 90 degree field of view, 21,600 effective points/s, an 11 Hz horizontal scan rate, a 250 Hz IMU stream, and +/- 2 cm measurement accuracy. No independent, controlled planar-target study was found that validates the +/- 2 cm claim across range, reflectivity, and incidence angle. Independent evidence instead concerns application-level mapping. Re-analysis of six CERN sequences shows that a scan-context-aware LiDAR-only method achieved an unweighted mean trajectory RMSE of 0.261 m relative to a Point-LIO reference, versus 3.532 m for Small GICP and 2.839 m for KISS-ICP. The corresponding mean reductions were 92.6% and 90.8%, although Small GICP was marginally better in one sequence. Field mapping in a lava tube produced coherent global morphology but insufficient density for detailed floor reconstruction. The evidence therefore supports conditional suitability: the L1 is credible for navigation-scale indoor perception when motion is moderate, overlap is preserved or reconstructed, and an LIO or non-repetitive-scan-aware registration method is used. It is not independently validated as a +/- 2 cm metrology instrument, is vulnerable to sparse overlap and ambiguous geometry with conventional ICP, and should not be used alone for fine surface reconstruction. Long-duration ROS 2 stability and coverage-versus-time remain unreported and require direct experiment.

DIRECT ANSWER

The Unitree L1 is conditionally adequate for indoor navigation and coarse 3D mapping, but the published record does not justify a general claim of +/- 2 cm field accuracy, guaranteed real-time mapping, or detailed geometric reconstruction. Reliability depends more strongly on scan overlap, motion, scene geometry, and the registration/fusion method than on the nominal range specification alone.

1. Research question and analytical scope

The target protocol formulated four hypotheses: H1, manufacturer-compatible range accuracy on a frontal matte target; H2, non-uniform spatial sampling that improves with accumulation; H3, independently measurable ROS 2 timing stability; and H4, repeatable navigation-scale mapping despite limited fine detail. The present work evaluates how far existing public evidence can answer each hypothesis.

This is a targeted evidence synthesis, not a systematic review or a replacement for the planned Coimbra measurements. Sources were included when they (i) explicitly identified the Unitree L1, (ii) were a manufacturer document, peer-reviewed primary study, or public primary dataset, and (iii) reported specifications, quantitative results, or direct field observations relevant to range, sampling density, stream behavior, odometry, or mapping. Search and access were completed on 6 August 2026.

2. Materials and methods

2.1 Evidence base

Source	Evidence used	Role in this study	Main limitation
Unitree manual [1]	Specifications and stated test conditions	Instrument envelope	Manufacturer evidence; no raw validation data
CERN study [2] and dataset [3]	Six L1 sequences; RMSE and runtime tables	Quantitative secondary analysis	Point-LIO is a sensor-fusion pseudo-reference, not external ground truth
Lava-tube field study [4]	Point counts, drift observations, map-use assessment	Independent field evidence on density and reconstruction	Uncontrolled cross-modality comparison; no 3D ground truth
Indoor hallway study [5]	Reported failure of an L1/Point-LIO comparator in two tests	Corroborating failure evidence	Full experimental table was not accessible; excluded from pooling
Livox MID-360 specifications [6]	200,000 points/s	Rate-scale comparison only	Different sensor geometry and architecture; not an accuracy benchmark

2.2 Secondary calculations

The six environment-level RMSE values in Table 3 of Habibiroudkenar et al. [2] were transcribed and checked against the published table. For each method, the unweighted arithmetic mean and median were calculated across environments. Relative improvement was calculated as $100 \times (1 - \text{mean RMSE}_{\text{method}} / \text{mean RMSE}_{\text{baseline}})$. These statistics are descriptive: sequence duration and sample counts were unavailable, so no pooled trajectory-level estimate or confidence interval was calculated.

The nominal processing budget corresponding to the L1 horizontal rate is $1/11 \text{ s} = 90.9 \text{ ms}$ per scan. A normalized point-rate comparison used 21,600 effective points/s for the L1 and 200,000 points/s for the MID-360. This ratio is not a direct measure of angular density because the sensors have different fields of view and scanning trajectories.

3. Results

3.1 Instrument envelope and measurement accuracy

Quantity	Published L1 value	Evidence status	Scientific interpretation
Range	PM: 20 m at 90% / 10 m at 10%; RM: 30 m at 90% / 15 m at 10%	Manufacturer only	Reflectivity materially changes usable range.
Accuracy	+/- 2.0 cm	Manufacturer only	Not independently verified across range or incidence.
Resolution	8 mm	Manufacturer only	Resolution must not be interpreted as absolute accuracy.
Field of view	360 x 90 degrees	Manufacturer; repeated in field studies	Very broad awareness, but not uniform point density.
Point rate	21,600 effective points/s	Manufacturer	About 1,964 points per 11 Hz horizontal cycle as a rate normalization.
IMU output	250 Hz	Manufacturer	Enables LIO, but timing quality and bias stability are not reported.

Finding 1. H1 cannot be confirmed from public independent evidence. The manual states +/- 2 cm and notes test conditions involving 25 degrees C, 90% reflectivity, and 15 m, but no public point-to-plane dataset was located that separates bias, repeatability, incidence angle, reflectivity, and reference-instrument uncertainty. The correct scientific conclusion is therefore "not independently validated," not "inaccurate."

3.2 Spatial sampling and coverage

The L1 uses a non-repetitive brushless-mirror pattern. At 21,600 effective points/s and 11 Hz, the rate corresponds to approximately 1,964 effective returns per horizontal cycle. The MID-360 publishes 200,000 points/s, about 9.26 times the L1 rate [6]. This comparison supports the CERN authors' observation that the L1 produces considerably fewer points and lower consecutive-scan overlap than the MID-360 [2], but it does not establish a universal density ratio because angular allocation differs.

Independent field evidence is consistent with limited fine-scale density. In the lava-tube study, the selected L1 floor cloud contained 656,889 points, while the photogrammetric floor model contained 1,260,001 points; the L1 cloud was 52.1% of that count under the mission-specific acquisition settings. The authors found the LiDAR adequate for overall morphology but insufficient for detailed floor conditions [4]. The comparison is observational rather than controlled, because scan spacing, exposure time, modality, and processing differ.

Finding 2. H2 is qualitatively supported: low and irregular overlap is a recurring limitation, and accumulation or deliberate stationary scanning improves usable density. However, no published L1 coverage-versus-time curve was found, so the time required to reach a specified angular coverage remains unknown.

3.3 Odometry and mapping performance

The strongest quantitative L1 evidence is the six-environment CERN dataset. The environments covered 5-20 m² floor areas with 3-7 m ceilings and included dynamic clutter, repetitive structures, tunnels, and a symmetric dome [2]. The published "ground truth" was generated with Point-LIO using the same L1 and its IMU; consequently, the results measure agreement with a strong fusion estimate rather than absolute trajectory error against motion capture or a total station.

Environment	Spatial-context method	Small GICP	KISS-ICP
CHARM	0.722	2.629	1.359
Beam Dump	0.037	2.920	0.807
BA80a	0.162	5.292	7.377
BA5	0.147	7.025	1.254
BA6	0.126	2.972	2.751
BA80b	0.373	0.356	3.486
Mean	0.261	3.532	2.839
Median	0.154	2.946	2.055

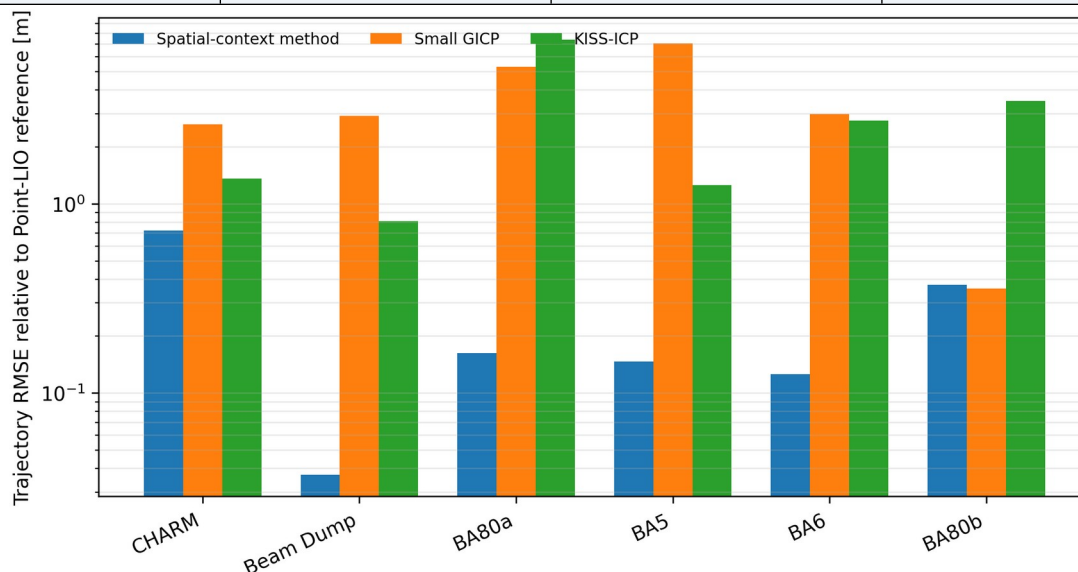


Figure 1. Published trajectory RMSE values for six Unitree L1 CERN sequences [2]. The logarithmic scale exposes both low-error and failure regimes. The reference trajectory was produced by Point-LIO, not an external metrology system.

Secondary-analysis result. The spatial-context method achieved a mean RMSE of 0.261 m and a median of 0.154 m. The unweighted mean was 92.6% lower than Small GICP and 90.8% lower than KISS-ICP. It was better than KISS-ICP in all six environments and better than Small GICP in five. In BA80b, Small GICP was marginally lower (0.356 m versus 0.373 m), so the published table does not support a literal claim of superiority in every environment.

The BA6 ablation provides an additional comparison: RMSE was 0.126 m for the proposed method, 0.349 m for kinematic ICP, and 0.346 m for wheel odometry. These correspond to reductions of 63.9% and 63.6%, respectively. The result supports the need for robust non-repetitive-scan registration, but it remains conditional on the Point-LIO reference and a single sequence.

3.4 Computational feasibility and ROS 2 stability

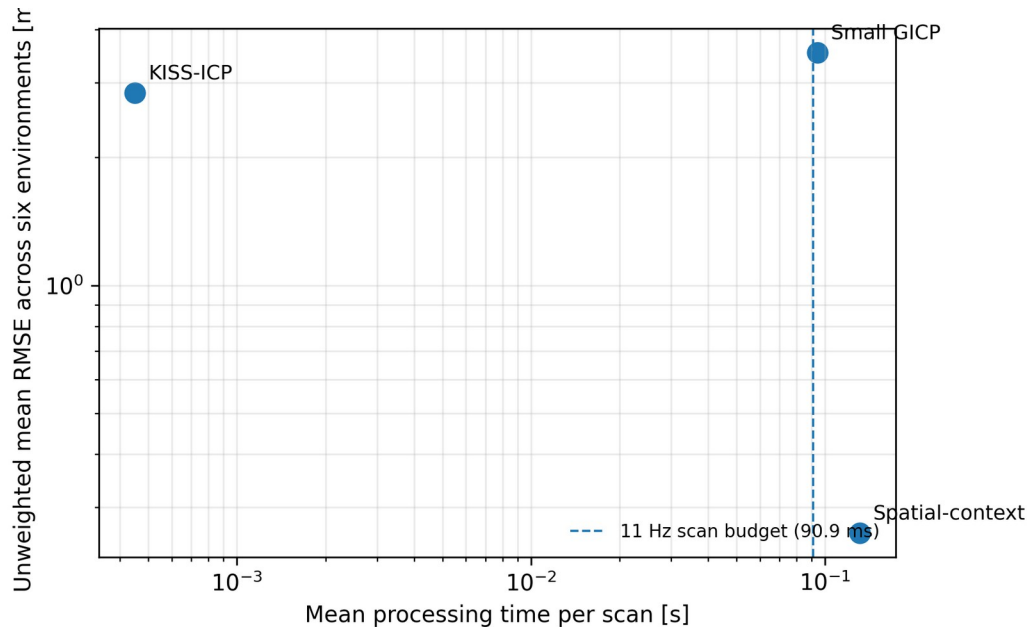


Figure 2. Accuracy-runtime trade-off derived from the six CERN sequences [2]. Each point is an unweighted environment mean. Both axes are logarithmic. The vertical line is the 90.9 ms budget implied by an 11 Hz cycle.

The spatial-context method required 0.131 s/scan on average, equivalent to about 7.6 scans/s. This is 44.1% above the 90.9 ms budget implied by 11 Hz, and only one of the six reported environment means was below that budget. The authors explicitly describe the evaluation as offline; the results therefore motivate, but do not establish, end-to-end real-time operation [2]. Small GICP averaged 0.094 s/scan, while KISS-ICP was much faster but substantially less accurate on these sequences.

Finding 3. H3 remains unanswered. Public demonstrations and datasets show that the L1 can feed ROS/Point-LIO pipelines, but no controlled 30-minute report was found for cloud and IMU frequency, timestamp jitter, packet loss, invalid returns, CPU load, or thermal behavior. Algorithm runtime must not be confused with sensor-stream stability.

4. Scientific interpretation

4.1 Conditions under which the L1 is supported

Condition	Evidence-based interpretation
Navigation-scale objective	Evidence supports room, corridor, tunnel, and cave morphology; it does not support fine surface metrology.
Moderate platform motion	Lower motion per update preserves overlap; early motion and reduced overlap were identified as difficult regimes.
Structured or sufficiently observed geometry	Walls, corners, ceilings, and repeated observations improve constraints; symmetry and repetitive corridors require context-aware registration.
LIO or scan-pattern-aware registration	Point-LIO provides a practical fused reference; a tailored LiDAR-only front end greatly outperformed generic ICP baselines.
Accumulation or deliberate pauses when detail matters	The lava-tube mission used scanning positions approximately 2-4 m apart

	and post-registration; stationary acquisition increased local density.
Operation within the rated reflectivity-dependent envelope	Range falls from 20/30 m at 90% reflectivity to 10/15 m at 10% for PM/RM variants; independent accuracy is still unverified.

4.2 Failure and caution regimes

- Fine floor roughness, thin-object reconstruction, and detailed geometry: the available field comparison shows insufficient density under realistic acquisition constraints.
- Raw scan-to-scan ICP in sparse, repetitive, symmetric, or low-overlap scenes: published errors reached multi-metre RMSE in several CERN sequences.
- High-speed or abrupt motion without reliable deskewing and inertial fusion: reduced consecutive overlap makes initialization and early trajectory estimation fragile.
- Claims of centimetric field accuracy: the ± 2 cm figure remains a manufacturer specification, not an independently reproduced uncertainty statement.
- Claims of real-time end-to-end performance: the best L1-specific method was evaluated offline and exceeded the nominal 11 Hz per-scan budget on average.
- Transparent, dark, strongly oblique, or thermally adverse conditions: the datasheet itself indicates reflectivity and environmental dependence, but independent L1-specific error curves are absent.

4.3 Hypothesis verdicts

Hypothesis	Verdict	Confidence	Reason
H1: ± 2 cm compatible range accuracy	Not established	Low	No independent controlled range-error dataset was found.
H2: non-uniform coverage improves with accumulation	Qualitatively supported	Moderate	Sparse overlap and density limitations are independently observed; no time-to-coverage curve exists.
H3: ROS 2 stream stability is characterisable	Plausible but unreported	Low	No long-duration timing, dropout, invalid-point, or thermal study was found.
H4: repeatable navigation-scale mapping	Conditionally supported	Moderate	Strong results require LIO or scan-aware registration; generic ICP frequently fails and fine detail is limited.

5. Limitations

- The literature is small and heterogeneous. Only one peer-reviewed study provided a full L1-specific odometry error table suitable for secondary analysis.
- The CERN reference trajectory was generated by Point-LIO from the same LiDAR plus IMU. It is a reasonable operational reference, but not independent ground truth; absolute RMSE should therefore be interpreted cautiously.
- Environment-level means were treated equally because sequence lengths and trajectory-sample counts were not reported. The calculated mean reductions are descriptive, not a formal meta-analysis.
- The lava-tube point-count comparison mixes LiDAR and photogrammetry under different acquisition and processing conditions. It supports a task-level density limitation but cannot estimate intrinsic sensor resolution.
- Raw CERN bag files are publicly available [3], but the present document re-analyses published tables rather than replaying the bags. A reproducible reprocessing study could test alternative LIO, filtering, and map-quality metrics.
- No evidence located by the search answers the planned planar-target, coverage-convergence, or 30-minute ROS 2 stability experiments. Those measurements remain scientifically useful and non-redundant.

6. Conclusion

The published evidence gives a conditional, application-level answer to the research question. The Unitree L1 can provide sufficiently stable spatial information for indoor mobile-robot perception and navigation-scale mapping when the robot moves at moderate speed, the scene supplies repeated geometric constraints, and processing explicitly accounts for the non-repetitive scan through LiDAR-inertial fusion, temporal accumulation, or context-aware registration. Under these conditions, the best published L1-specific LiDAR-only method remained below 0.722 m RMSE relative to a Point-LIO reference in all six CERN environments and achieved a 0.261 m unweighted mean.

The evidence does not support three stronger claims: that field measurements are generally accurate to ± 2 cm, that raw L1 scans are reliably handled by conventional ICP, or that the sensor provides enough density for detailed geometric reconstruction. The absence of independent range metrology, coverage-versus-time curves, and long-duration ROS 2 diagnostics is not a reason to abandon the Coimbra protocol; it is precisely the scientific gap that makes those experiments valuable.

OPERATIONAL RECOMMENDATION

Use the L1 as a low-cost, wide-field navigation sensor, not as a calibrated survey instrument. Pair it with its IMU and a validated LIO pipeline; accumulate scans or pause for local detail; monitor overlap and initialization; and retain the planned planar-target and ROS 2 stability tests before making centimetric or real-time performance claims.

Data and reproducibility statement

The CERN Unitree L1 bag files and IMU data are publicly archived on Zenodo under DOI 10.5281/zenodo.17091989 [3]. The numeric values used in Figures 1 and 2 and in the RMSE table were transcribed from Tables 3 and 5 of Ref. [2]. Derived calculations are fully specified in Section 2.2 and can be reproduced from the six RMSE and runtime values printed in this document.

References

- [1] Unitree Robotics, Unitree 4D LiDAR-L1 User Manual, Version 1.1, June 2024. Available: <https://www.unitree.com/download/LiDAR/> (accessed 6 August 2026).
- [2] P. Habibiroudkenar, D. Forkel, R. Ojala, S. M. Hari Prasanth, D. H. Davidsson, K. Tammi, E. Matheson, and M. Di Castro, "Non-Repetitive Scan Correction for LiDAR Odometry via Spatial Analysis in CERN Facilities," *IEEE Access*, vol. 14, pp. 52717-52733, 2026, doi: 10.1109/ACCESS.2026.3680632.
- [3] P. Habibiroudkenar, D. Forkel, and H. D. Dadi, "Unitree Unilidar L1 scans in CERN facilities," Zenodo, 10 September 2025, doi: 10.5281/zenodo.17091989.
- [4] A. J. Hidding, H. H. Bier, L. Peternel, A. J. Becoy, F. A. P. Romio, and G. Calabrese, "Robotic lava tube mapping and multimodal data collection using quadruped and LiDAR," *Frontiers in Robotics and AI*, vol. 13, art. 1855443, 2026, doi: 10.3389/frobt.2026.1855443.
- [5] H. Cheng and J. Han, "Toward precise dense 3D reconstruction of indoor hallway: a confidence-based panoramic LiDAR point cloud fusion approach," *Industrial Robot*, vol. 52, no. 1, pp. 116-125, 2025, doi: 10.1108/IR-03-2024-0132.
- [6] Livox Technology, "MID-360 LiDAR Sensor - Specifications," manufacturer technical page. Available: <https://www.livoxtech.com/mid-360/specs> (accessed 6 August 2026).
- [7] D. He, W. Xu, N. Chen, F. Kong, C. Yuan, and F. Zhang, "Point-LIO: Robust High-Bandwidth Light Detection and Ranging Inertial Odometry," *Advanced Intelligent Systems*, vol. 5, no. 7, art. 2200459, 2023, doi: 10.1002/aisy.202200459.
- [8] Unitree Robotics, "point_lio_unilidar: Point-LIO algorithm adapted for Unitree LiDAR products," public software repository. Available: https://github.com/unitreerobotics/point_lio_unilidar (accessed 6 August 2026).